P6 | Duplex seawater pump

Water-lubricated axial-piston technology, selected for saline-fluid duty.

Duplex construction supports the seawater application; it does not replace a materials review. Specify salinity, temperature, dissolved solids, cleaning chemicals and standby/flushing conditions along with flow and pressure.

Published parameter	4.6 cm³/rev	6.0 cm³/rev
Maximum boosted speed (rpm)	2000	2000
Maximum input power (kW)	1.4	1.8
Maximum flow (m³/h)	0.55	0.72
Maximum continuous pressure (bar)	80	80
Mass (kg)	2.2	2.2
Maximum temperature (°C)	50	50
Minimum temperature (°C)	2	2

Flow equivalents (US gpm): 2.42, 3.17. 80 bar = 1,160 psi; 2-50 °C = 35.6-122 °F. Each column is a displacement option; maxima are independent ratings.

Design & environmental advantages

Product-water lubrication removes the pump oil circuit and its contamination pathway. A compact multi-piston arrangement produces smooth flow. Duplex materials address saline duty more deliberately than a generic freshwater specification.

Efficiency at the operating point

Compare input power with useful hydraulic output at the same displacement, speed and pressure. The supplied flow, power, torque and efficiency charts identify those conditions; they are not interchangeable with the 160 bar freshwater range.

Feed, drainage and installation

Published feed-pressure guide: 3.0-3.6 bar (44-52 psi), depending on shaft speed. Maintain inlet pressure at least 2.5 bar (36 psi) above case pressure. Separate internal leakage to a low-pressure drain. Specify coupling, rotation, supports, filtration, start-up flushing and service access.

Quotation and technical submittal

Provide flow, discharge pressure, inlet pressure at the pump, speed, feed analysis, duty cycle and site conditions. Request the selected displacement, wetted materials and seals, approved drawing, drive selection, pressure-test records and maintenance schedule. Match corrosion claims to the actual chemistry.

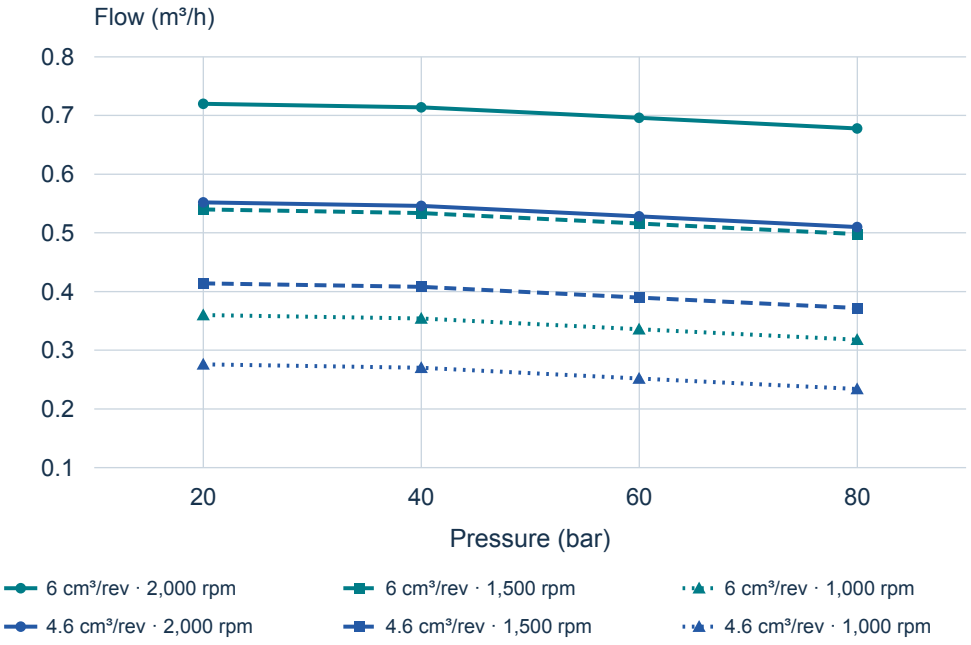
No oil in the pump circuit reduces one contamination risk. Brine handling, electricity use, water quality and the rest of the installation remain part of the environmental assessment.

Sea Water Axial Piston Pumps V.04.26 (3).pdf, pp. 2-4, 7-9.

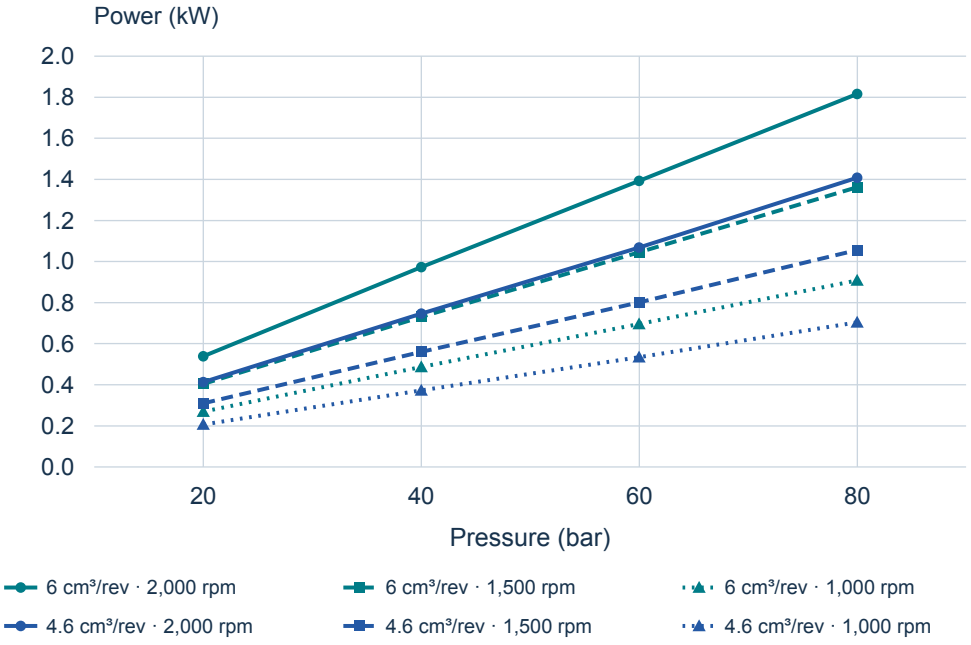
P6 | Performance

Read the source legend and stated operating conditions for each chart.

Output flow



Input power



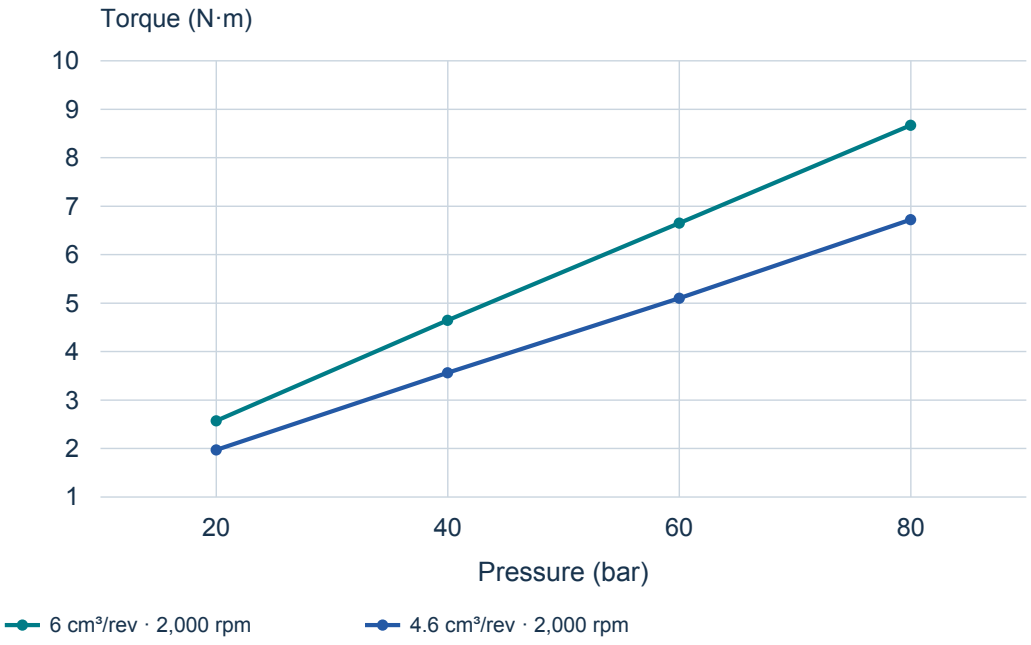
Colours identify displacement; line patterns and markers distinguish shaft speed. Native curve geometry was checked against the manufacturer PDF. 1 m³/h = 4.40287 US gpm; 1 N·m = 0.737562 lbf·ft.

Sea Water Axial Piston Pumps V.04.26 (3).pdf, pp. 8, 9.

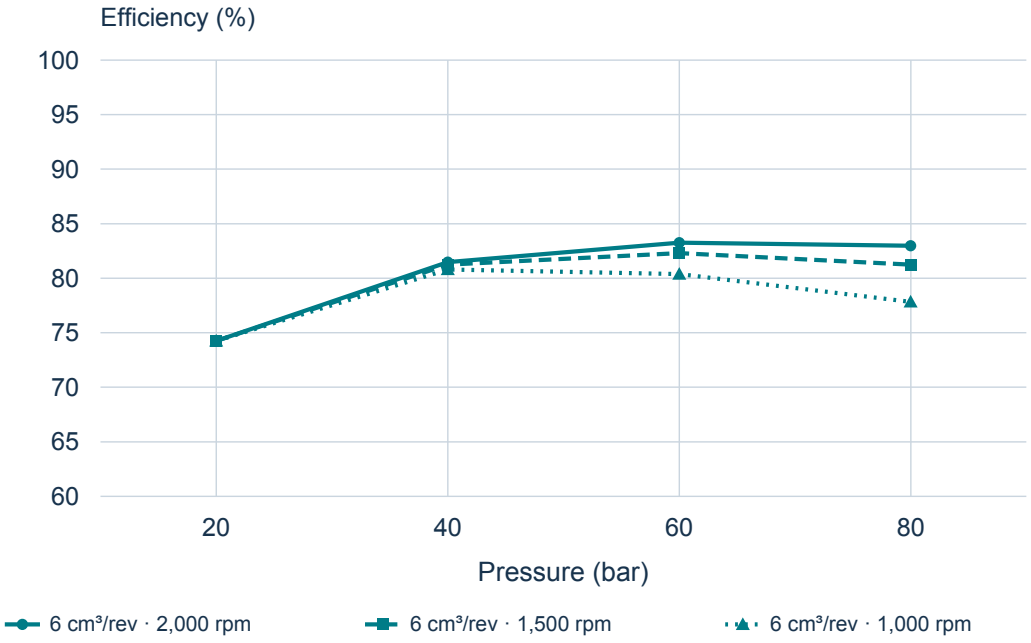
P6 | Performance

Read the source legend and stated operating conditions for each chart.

Input torque



Overall efficiency



Colours identify displacement; line patterns and markers distinguish shaft speed. Native curve geometry was checked against the manufacturer PDF. 1 m³/h = 4.40287 US gpm; 1 N·m = 0.737562 lbf·ft.

Sea Water Axial Piston Pumps V.04.26 (3).pdf, pp. 8, 9.

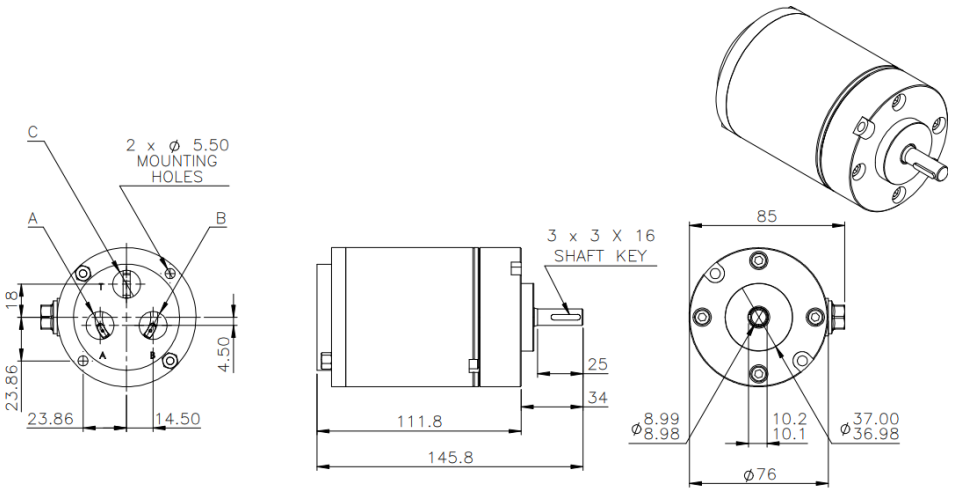
P6 | Dimensions & ports

Manufacturer technical reference | dimensions govern; reproduction is not to scale.

3.1.1 Specification

SPECIFICATION			
Displacement (cc/rev)		4.6	6.0
Max. RPM	Boosted: *	2000	2000
Max. Input Power (kW)		1.4	1.8
Max. Water Flow (m³/hr)		0.55	0.72
Max. Cont. Pressure (Barg)		80	80
Weight (kg)		2.2	2.2
Temperature (°C)	Max: **	50	50
	Min: ***	2	2

3.1.2 General Assembly Drawing



3.1.3 Port Dimensions (ACW Configuration)

Position	Description	Size
A	Low Pressure In	3/8” BSPP
B	High Pressure Out	3/8” BSPP
C	Tank Return Line	3/8” BSPP

The manufacturer specification and drawing are reproduced together to preserve port labels and configuration context. Nominal connection and flange ratings do not change the 80 bar pump operating limit.

Sea Water Axial Piston Pumps V.04.26 (3).pdf, p. 7.